

2. Why Do We Need Design Patterns?

2.1 The Reality of Software Development

- Ask students:

→ “Why do large software systems become difficult to maintain over time?”

→ Common answers:

- Code becomes complex
- Changes break existing features
- Adding new features takes too long
- New developers struggle to understand the code

→ Explain:

- These problems do not come from syntax or programming languages — they come from poor design decisions.